

**Multiple Choice:**

1. A car is traveling at 20 m/s away from a stationary observer. If the car's horn emits a frequency of 600 Hz, what frequency will the observer hear? (Use $v = 340$ m/s for the speed of sound.)
A. $(34/36)(600 \text{ Hz})$
B. $(34/32)(600 \text{ Hz})$
C. $(36/34)(600 \text{ Hz})$
D. $(32/34)(600 \text{ Hz})$
2. An observer is 2 m from a source of sound waves. By how much will the sound level decrease if the observer moves to a distance of 20 m?
A. 1 dB
B. 18 dB
C. 10 dB
D. 20 dB
3. A sound wave traveling through water has a frequency of 500 Hz and a wavelength of 3 m. How fast does sound travel through water in m/s?
A. 1000
B. 1500
C. 950
D. 750
4. A sound wave with a frequency of 343 Hz travels through the air. What is its wavelength?
A. 1 m
B. 0.1 m
C. 1.2 m
D. 5 m
5. A transverse wave on a long horizontal rope with a wavelength of 8 m travels at 2 m/s. At $t = 0$, a particular point on the rope has a vertical displacement of $+A$, where A is the amplitude of the wave. At what time will the vertical displacement of this same point on the rope be $-A$?
A. $t = s/8$
B. $t = s/2$
C. $t = 2s$
D. $t = s/4$
6. A horizontal rope with linear mass density $\mu = 0.5 \text{ kg/m}$ has a tension of 50 N. The nonattached end is oscillated vertically with a frequency of 2 Hz. What is the wavelength of the resulting wave?
A. 0.05 m
B. 0.5 m
C. 5 m
D. 50 m
7. A rope of length 5 m is stretched to a tension of 80 N. If its mass is 1 kg, at what speed would a 10 Hz transverse wave travel down the string?
A. 2 m/s
B. 4 m/s
C. 20 m/s
D. 40 m/s
8. An object of height 4 cm is placed 30 cm in front of a concave mirror whose focal length is 10 cm. What is the height of the image?
A. 4 cm, upright
B. 4 cm, inverted
C. 6 cm, upright
D. 6 cm, inverted
9. When green light (wavelength = 500 nm in air) travels through diamond (refractive index = 2.4), what is its wavelength?
A. 208 nm
B. 500 nm
C. 357 nm
D. 700 nm
10. An object is placed 60 cm in front of a concave spherical mirror whose focal length is 40 cm. Which of the following best describes the image?
A. Image is virtual, 24 cm away
B. Image is real, 24 cm away
C. Image is virtual, 120 cm away
D. Image is real, 120 cm away
11. What is the wavelength of an X-ray whose frequency is $1.0 \times 10^{18} \text{ Hz}$?
A. $3.3 \times 10^{-11} \text{ Hz}$
B. $3.0 \times 10^{-10} \text{ Hz}$
C. $3.3 \times 10^{-9} \text{ Hz}$
D. $3.3 \times 10^{25} \text{ Hz}$
12. A concave mirror with a focal length of 25 cm is used to create a real image that has twice the height of the object. How far is the image from the mirror?
A. 45 cm
B. 60 cm
C. 75 cm
D. 90 cm
13. An object of height 10 cm is placed 40 cm in front of a converging lens with a focal length of 20 cm. Where's the image?
A. 20 cm
B. 30 cm
C. 40 cm
D. 50 cm
14. Which of the following is true concerning a bi-concave lens?
A. Its focal length is positive.
B. It cannot form real images.
C. It cannot form virtual images.
D. It can magnify objects.
15. A beam of light in air is incident upon a piece of glass, striking the surface at an angle of 60° . If the index of refraction of the glass is 1.5, what are the angles of reflection and refraction, respectively?
A. 60° and 19°
B. 30° and 19°
C. 60° and 38°
D. 30° and 38°
16. Combinations of lenses are often used. The image of the first lens serves as the object for the second lens. Suppose two thin coaxial lenses with focal lengths $f_1 = 25 \text{ cm}$ and $f_2 = 10 \text{ cm}$ are separated by 8 cm. An insect is positioned 6 cm from lens 1 (on the opposite side from lens 2). Where is its final image?
A. 27 cm from the second lens
B. 41 cm from the second lens
C. 34 cm from the second lens



- D. 21 cm from the second lens
17. How long will it take for sound to travel a distance of 4 km if the air temperature is increased from 0°C to 20°C?
- A. 9.5 s
B. 10.1 s
C. 11.7 s
D. 12.8 s
18. A choir is composed of 39 singers all of whom can sing with the same intensity. A soloist sings a passage alone and is joined by the rest for the repeat passage. Calculate the difference in sound intensity level in the two cases.
- A. 16 dB
B. 32 dB
C. -16 dB
D. -32 dB
19. What is the intensity level at a point 20 m from the source if a sound energy is radiating uniformly in all directions at a rate of 2 watts?
- A. 68 dB
B. 86 dB
C. -86 dB
D. -68 dB
20. A sound has an intensity of 10^{-10} watts/cm². What is its intensity level in dB?
- A. 60 dB
B. -60 dB
C. 80 dB
D. -80 dB
21. A human ear can distinguish a reflected sound pulse from the original sound if 0.10 s or more elapses between the two sounds. If an echo is heard exactly 0.20 s after a sound of 1150 ft/s is made, find the distance of the reflecting surface.
- A. 115 ft
B. 144 ft
C. 132 ft
D. 152 ft
22. A ray of light travels from air at an angle of 42° with the water surface. What is the speed of light in water (in m/s) if the index of refraction of water is 1.33?
- A. 2.13×10^8
B. 2.26×10^8
C. 3.13×10^8
D. 3.52×10^8
23. The speed of light in a particular substance is 123,000 miles/s. What is the refractive index of this substance?
- A. 1.35
B. 1.52
C. 2.12
D. 1.81
24. An incident ray from water makes an angle of 55° with respect to the normal. What is the direction of the reflected ray if it goes to glass with an index of refraction of 1.50?
- A. 55°
B. 52°
C. 59°
D. 62°
25. A diverging lens has a focal length of -5.0 cm. If you look through the lens at a dime held 4.0 cm from the lens, what is the magnification of the image?
- A. 0.55
B. -0.55
C. 1.55
D. -1.55
26. A far-sighted classmate is unable to clearly focus on objects closer than 125 cm from his eyes. What strength corrective lens is needed to allow him to read books held 30 cm from his eyes?
- A. 2.1 diopters.
B. 2.5 diopters
C. 3.4 diopters
D. 3.2 diopters
27. A common unit of energy is the electronvolt. 1 eV = ____ J.
- A. 1.8×10^{-19}
B. 1.6×10^{-19}
C. 1.4×10^{-19}
D. 1.2×10^{-19}
28. All are characteristics of waves EXCEPT:
- A. Energy of a wave varies with the square of its amplitude.
B. Interference is interaction of waves.
C. Diffraction/bending of a wave, occurs around the edges of obstacles.
D. Wave frequency changes on entering a medium where the velocity of propagation is changed.
29. The ratio of the speed of light in a vacuum to the speed of light in some transparent materials is called
- A. the critical angle
B. total internal reflection
C. the law of reflection
D. the index of reflection
30. A photon of which of the following has the most energy?
- A. Red light
B. Orange light
C. Green light
D. Blue light
31. The angle of incidence is the angle that an incident ray makes with the normal to the surface while the angle of reflection is the angle that the reflected ray makes with the normal to the surface. According to the Law of Reflection, what is the relationship between the two angles?
- A. Angle of incidence is always greater
B. Angle of reflection is always greater
C. Both angles are equal
D. It depends on the medium where the light or the wave is traveling
32. Light travels _____ in matter than in vacuum.
- A. faster
B. at the same speed
C. slower
D. straighter



33. A person standing in waist-deep water appears to be shorter. This optical illusion is brought about by ____.
- A. reflection C. diffraction
B. refraction D. magic
34. Which of the following best describes the range 4000 to 7500 angstrom?
- A. infrared
B. visible light
C. ultraviolet
D. X-ray
35. If light comes from a medium of low refractive index and transfers to a medium of high refractive index, describe the relationship between the angle of incidence and the angle of refraction.
- A. angle of incidence and angle of refraction are equal
B. angle of incidence is greater than the angle of refraction
C. angle of incidence is less than the angle of refraction
D. no relationship can be deduced from the statement
36. It refers to the spreading of white light into the different colors.
- A. diffraction C. refraction
B. dispersion D. prism
37. In a glass prism, which of the following light is bent the most?
- A. red
B. green
C. orange
D. violet
38. This phenomena is evident in making and designing plane and spherical mirrors.
- A. Reflection
B. Diffraction
C. Refraction
D. Interference
39. Which of the following mirrors has the widest field of view?
- A. Plane mirrors
B. Convex mirrors
C. Concave mirrors
D. Spherical mirrors
40. For its magnification properties, which of the following mirrors is commonly used as the large mirrors in astronomical telescopes?
- A. Plane mirrors
B. Convex mirrors
C. Concave mirrors
D. Spherical mirrors
41. These are very common mirrors used as “auxiliary rear-view mirrors” at the back of vans and SUVs.
- A. Plane mirrors
B. Convex mirrors
C. Concave mirrors
D. Spherical mirrors
42. It is a type of image that would not appear on paper or film placed at the location of the image.
- A. real image
B. virtual image
C. imaginary image
D. virtual object
43. Which of the following is NOT a characteristic of a plane mirror?
- A. Plane mirrors produce real images
B. Plane mirrors produce upright images
C. Plane mirrors has a magnification of +1
D. Plane mirrors produce “flipped” images
44. It is a type of mirror wherein the reflection takes place on the outer surface of the sphere.
- A. Convex
B. Spherical
C. Concave
D. Converging
45. Which type of mirror is commonly used in parking areas to provide a much wider view?
- A. Convex
B. Spherical
C. Concave
D. Plane
46. The focal length of a spherical mirror is ____ the radius of curvature.
- A. one-half
B. twice
C. equal
D. one-fourth
47. For concave mirrors, which of the following best describes the image formed if the object is located within the focal point?
- A. the image is upright
B. the image is real
C. the image is smaller
D. the image is inverted
48. For concave mirrors, what is the magnification if the object is located within the focal length?
- A. magnification is equal to 1
B. magnification is less than 1
C. magnification is greater than 1
D. magnification can be greater than or less than 1
49. For concave mirrors, what is the magnification if the object is located outside the focal length?
- A. magnification is equal to 1
B. magnification is less than 1
C. magnification is greater than 1
D. magnification can be greater than or less than 1
50. What is the sign of the focal length for convex spherical mirrors?
- A. positive
B. negative
C. positive or negative
D. can't be determined

END